Management Information Systems

Week 2

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Weekly Topics

- **Subject to change without prior notice**
- **Ad-hoc in-class writing or quiz and discussions will be given**

W	Ch	Topic	W	Ch	Topic
1		Orientation Introduction to MIS	9	7	Telecommunications, the internet, and wireless technology
2	**1**	**Information Systems in Global Business Today**	10	8	Securing information systems
3	2	Global E-business and Collaboration	11		*No Class*
4	3	Information Systems, Organizations, and strategy	12	9	Achieving operational excellence and customer intimacy
5	4	Ethical and Social issues in information systems	13	10	E-commerce: digital markets, digital goods
6	5	IT infrastructure and emerging technologies	14	11	Managing knowledge
7	6	Foundations of Business Intelligence	15	12/ 13	Enhancing decision making / Building information systems
8		*Midterm Exam*	*16*		*Final Exam*

Week 2

Information Systems in Global Business Today

Questions to Answer

1. How are information systems transforming business, and why are they so essential for running and managing a business today?

2. What is an information system? How does it work?
   What are its management, organization, and technology components?
   Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?

3. What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

1-1.

How are information systems transforming business?

&

Why are they so essential for running and managing a business today?

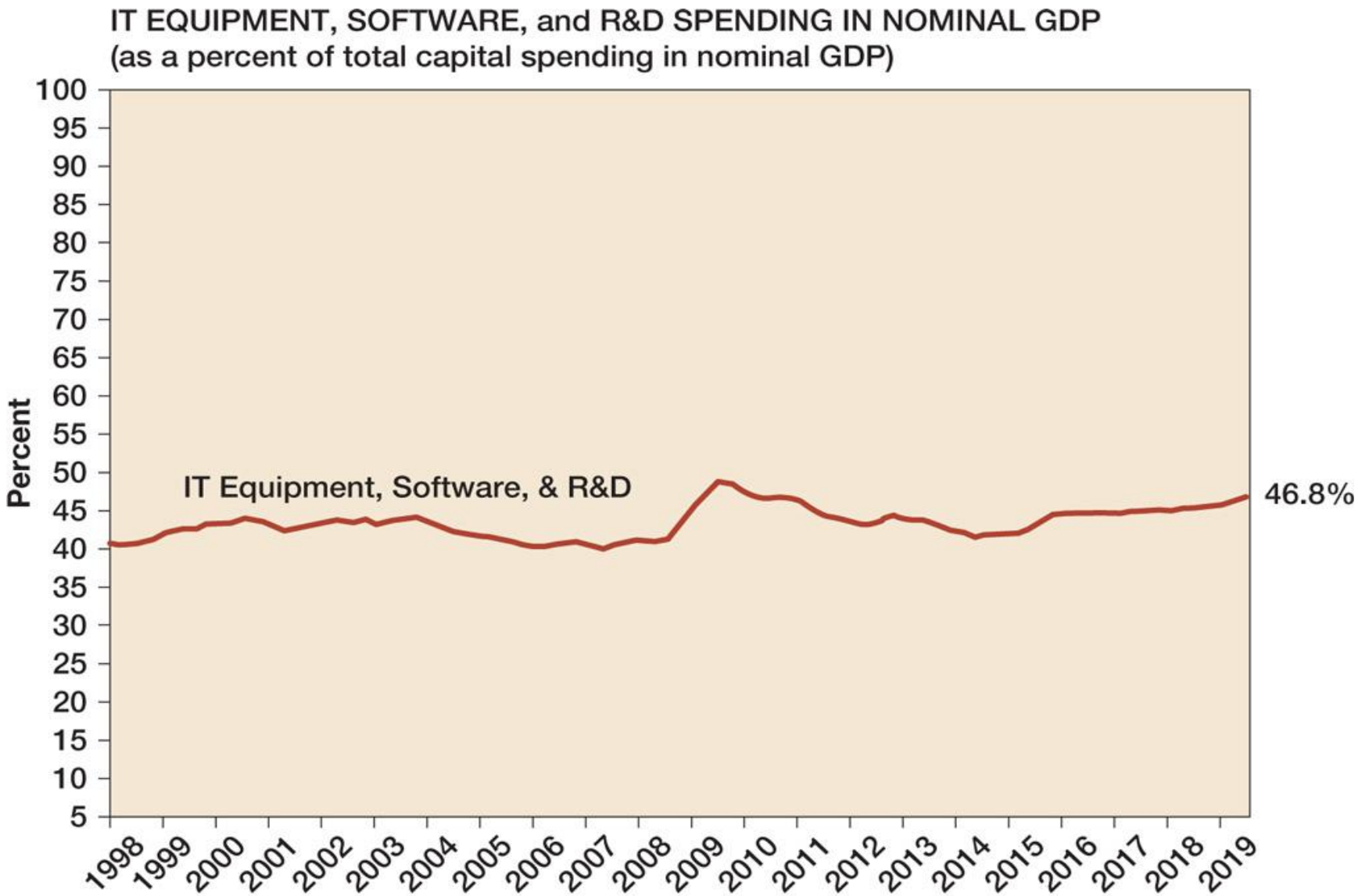
Introduction

- **Information systems and technologies are transforming the global business environment.**
  - Global spending on information technology (IT) and IT services: ~$3.8 trillion in 2019

- **Organizational, management, and cultural *changes* are often required for firms to derive full business value from IT investments**

- **This course aims to help you make *wise decisions* about information technology and information systems**

Information Technology Capital Investment



Source: US Bureau of Economic Analysis.

What's New In Management Information Systems

- IT Innovations
- New Business Models
- E-commerce Expanding
- Management Changes
- Changes in Firms and Organizations

E-commerce Expansion & Management Changes

- **E-commerce Expansion**
  - E-commerce worldwide expands to nearly $3.6 trillion in 2019
  - Growth in social commerce spurred by the growth of mobile platform
  - ***Mobile retail e-commerce*** is growing more than 20 percent a year, reaching almost $300 billion in 2020

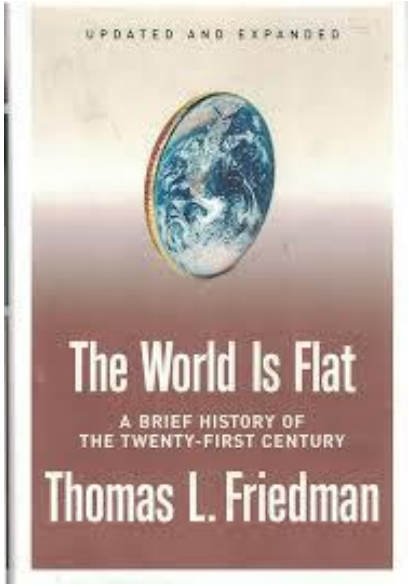
- **Management Changes**
  - Managers becoming more mobile
  - Managers use social networks, collaboration tools
  - Business intelligence applications accelerate

Changes in Firms and Organizations

- More collaborative, less emphasis on hierarchy and structure
- Greater emphasis on competencies and skills
- Higher-speed/more accurate decision-making based on ***data and analysis***
- More willingness to interact with consumers (social media)
- A better understanding of the importance of IT

Globalization Challenges and Opportunities: A Flattened World

- **Internet and global communications have significantly changed how and where business is done**
  - Customers can now shop in a worldwide marketplace, obtaining price and quality information reliably 24 hours a day.
  - Drastic ***reduction of costs*** of operating and transacting on a global scale
  - Competition for jobs, markets, resources, ideas
  - Growing interdependence of global economies



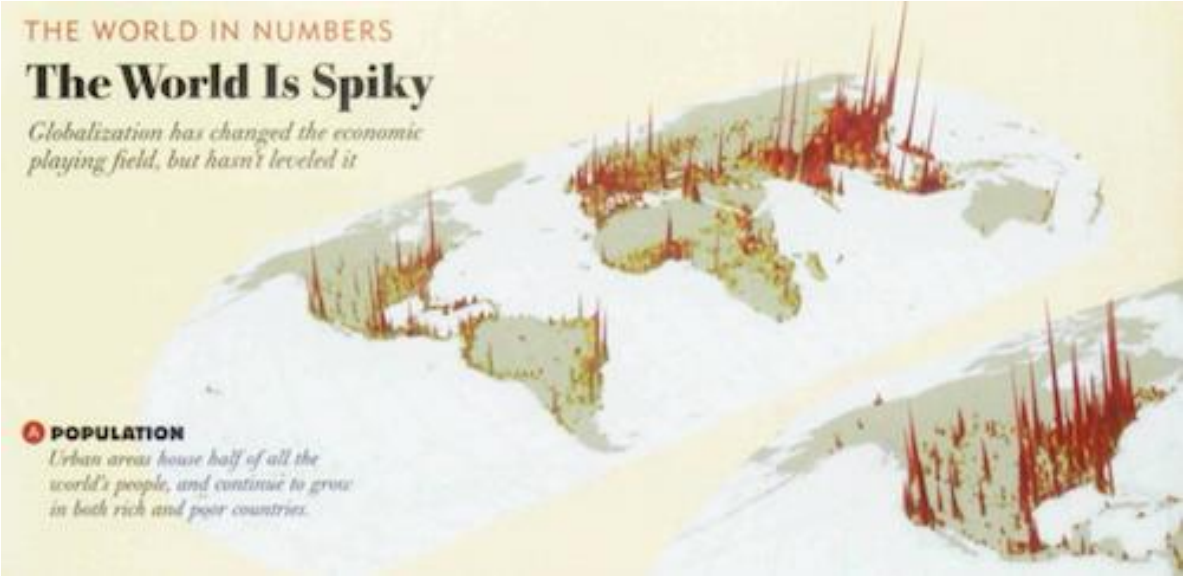


Image Source: Amazon, Medium

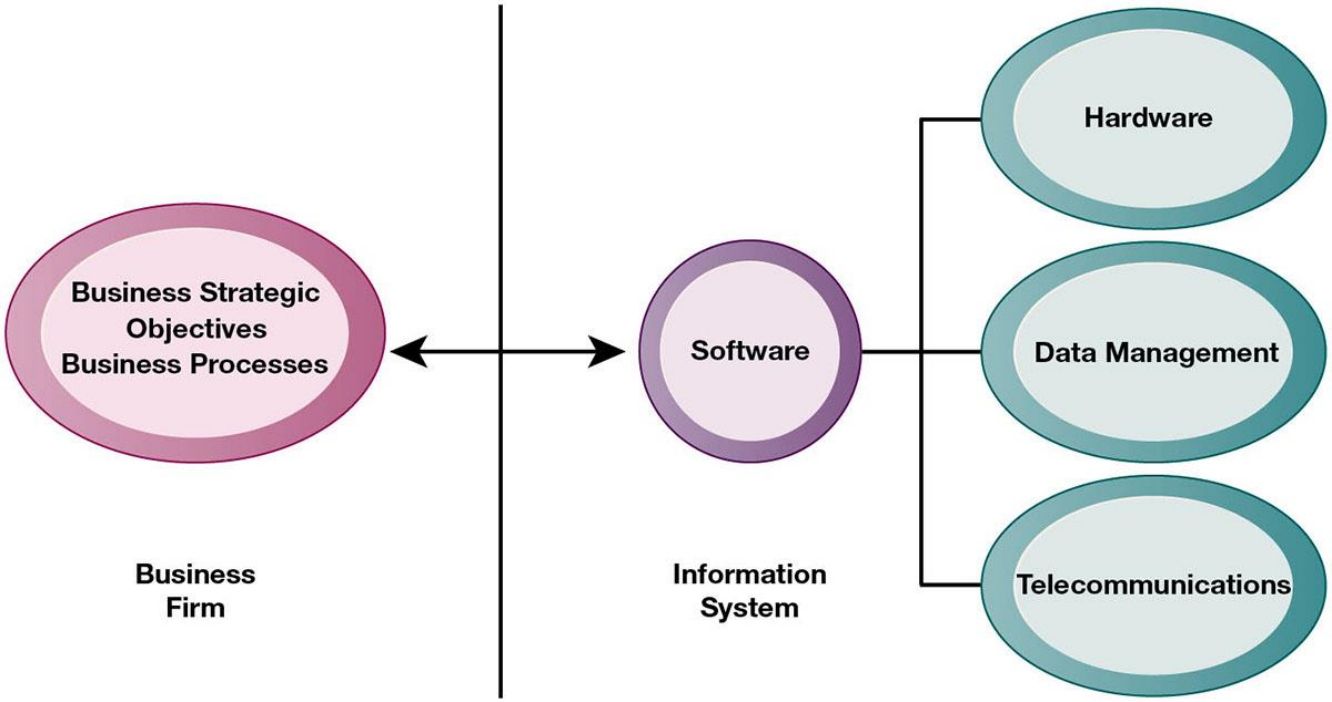
The Emerging Digital Firm

- **In a fully digital firm:**
  - Significant business relationships are ***digitally*** enabled and mediated
  - Core business processes are accomplished through ***digital*** networks
  - Key corporate assets (intellectual property, core competencies, and financial and human assets) are managed ***digitally***

- **Digital firms offer greater *flexibility* in organization and management**
  - Time shifting
  - Space shifting

Strategic Business Objectives of Information Systems

- **Growing interdependence between:**
  - Ability to use information technology
  - Ability to implement corporate ***strategies*** and achieve corporate goals
- **What a business would like to do in five years often depends on what its *systems* can do.**



The Interdependence Between Organizations and Information Systems

Strategic Business Objectives of Information Systems

- **Firms invest heavily in information systems to *achieve six strategic business objectives*:**

1. Operational excellence
2. New products, services, and business models
3. Customer and supplier intimacy
4. Improved decision making
5. Competitive advantage
6. Survival

Operational Excellence

1

- **Improvement of _____ to attain higher profitability**

- **Information systems and technology are some of the most important tools in achieving greater _____ and productivity**

- **Example: Walmart's Retail Link system for superior replenishment system**
  - Most efficient retail store in the world as a result of ***digital links*** between suppliers and stores

Image Source: 8th and Walton

New Products, Services, and Business Models

2

- **Information systems and technology are a major enabling tool for *new* products, services, business models**

- **Business model: describes how company produces, delivers, and sells product or service to create wealth**

- **Examples: Apple**
  - Transformed old model of music distribution with iTunes
  - Constant innovations - iPod, iPhone, iPad, etc.

Customer and Supplier Intimacy

3

- ***Customers*** **who are served well come back and purchase more**
  - Mandarin Oriental Hotel uses IT to foster an intimate relationship with customers, keeping track of preferences, etc.
- **Close relationships with *suppliers* result in lower costs**
  - JC Penney uses IT to enhance relationships with the supplier in Hong Kong.

Image Source: Wikipedia

Improved Decision making

4

- **Information systems and technologies have allowed managers to use *real-time data* from the marketplace when making decisions**

- **Without accurate information:**
  - Managers must use forecasts, best guesses, luck
  - Results in: *over/under-production*, misallocation of resources, poor response times ➔ Poor outcomes raise costs, lose customers

- **Verizon's web-based *digital dashboard* to provide managers with real-time data on customer complaints, network performance, etc.**

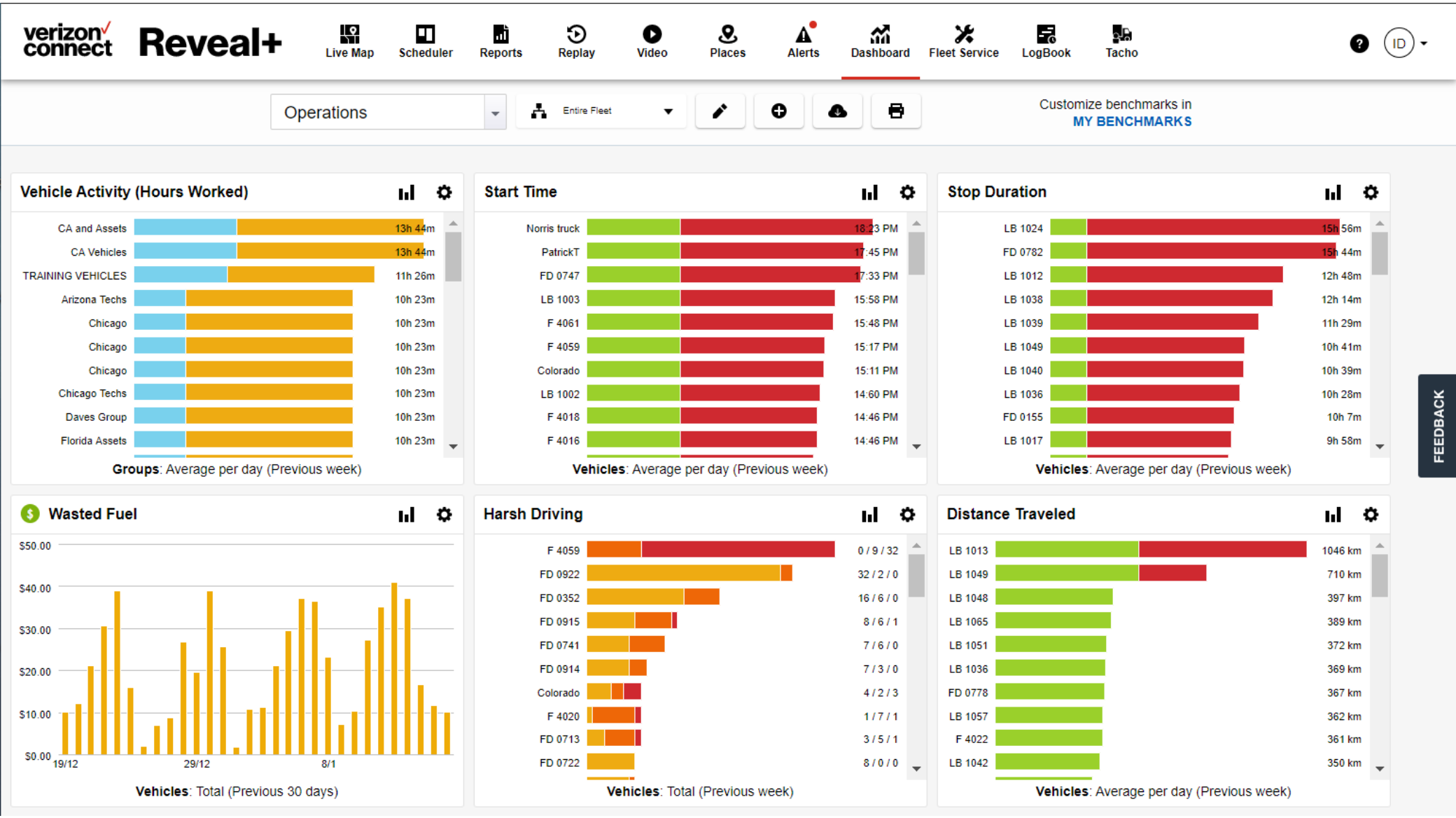


Image Source: https://reveal-help.verizonconnect.com/hc/en-us/articles/360010610359-Dashboard

Competitive Advantage & Survival

5 6

- **Competitive advantage: advantages over competitors**
  - Charging less for superior products, better performance, and better response to suppliers and customers
  - Apple, Walmart, UPS are industry leaders because they know how to use information systems for this purpose

- **To survive, businesses may need to invest in information systems out of necessity; it is simply the cost of doing business.**

Discussion

- **Do you agree with the statement below:**
  "In the AI era, information systems are no longer a source of competitive advantage; they are mainly a requirement for survival."

Firms invest heavily in information systems to *achieve six strategic business objectives*:

1. Operational excellence
2. New products, services, and business models
3. Customer and supplier intimacy
4. Improved decision making
5. Competitive advantage
6. Survival

➡ ?

1-2.

What is an information system?
How does it work?

What are its management, organization, and technology components?

Why are complementary assets essential for ensuring that IS provide genuine value for organizations?

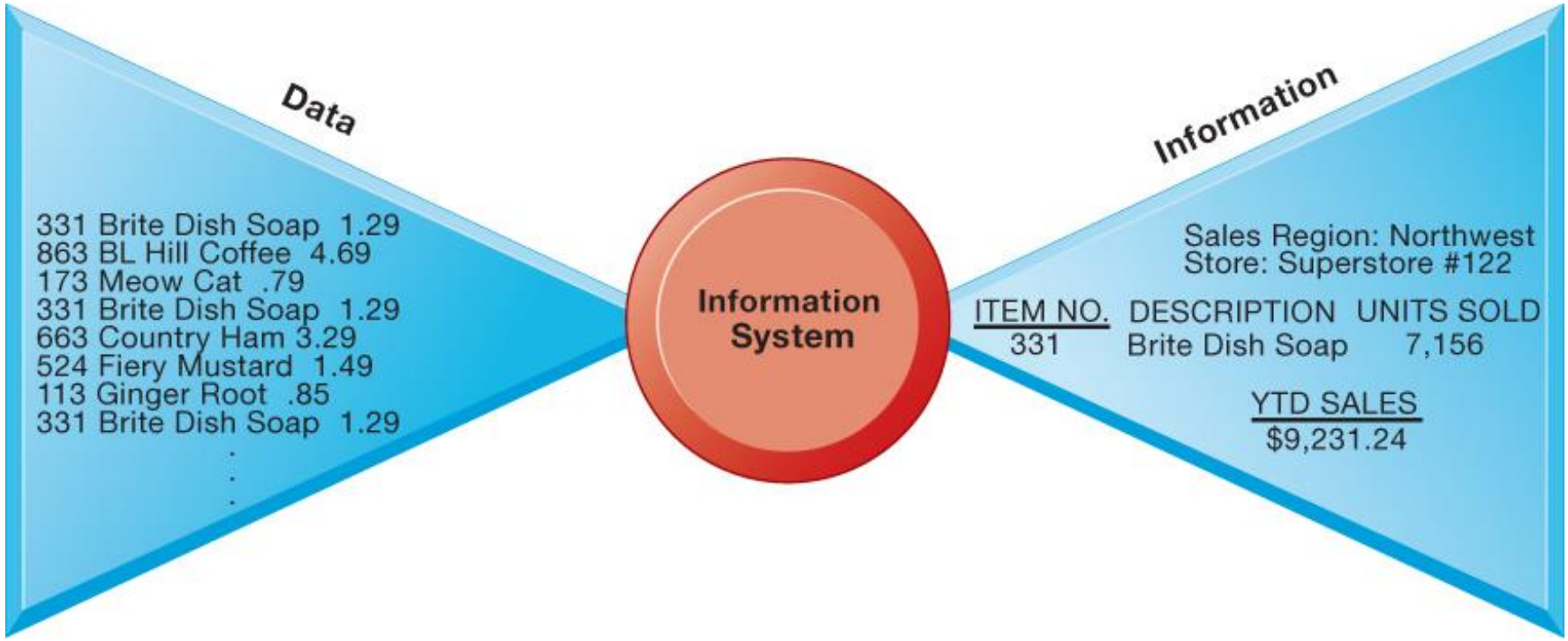
What Is an Information System?

- **Information technology (IT): hardware and software used to achieve business objectives**
  - HW: computer machines, storage devices, and handheld mobile devices, etc.
  - SW: operating systems, MS Offices, etc.

- **Information system**
  - Set of interrelated components
  - ***Collect, process, store, and distribute*** information to support decision-making, coordination, and control in an organization.
  - Help managers and workers analyze problems, visualize complex subjects, and create new products.

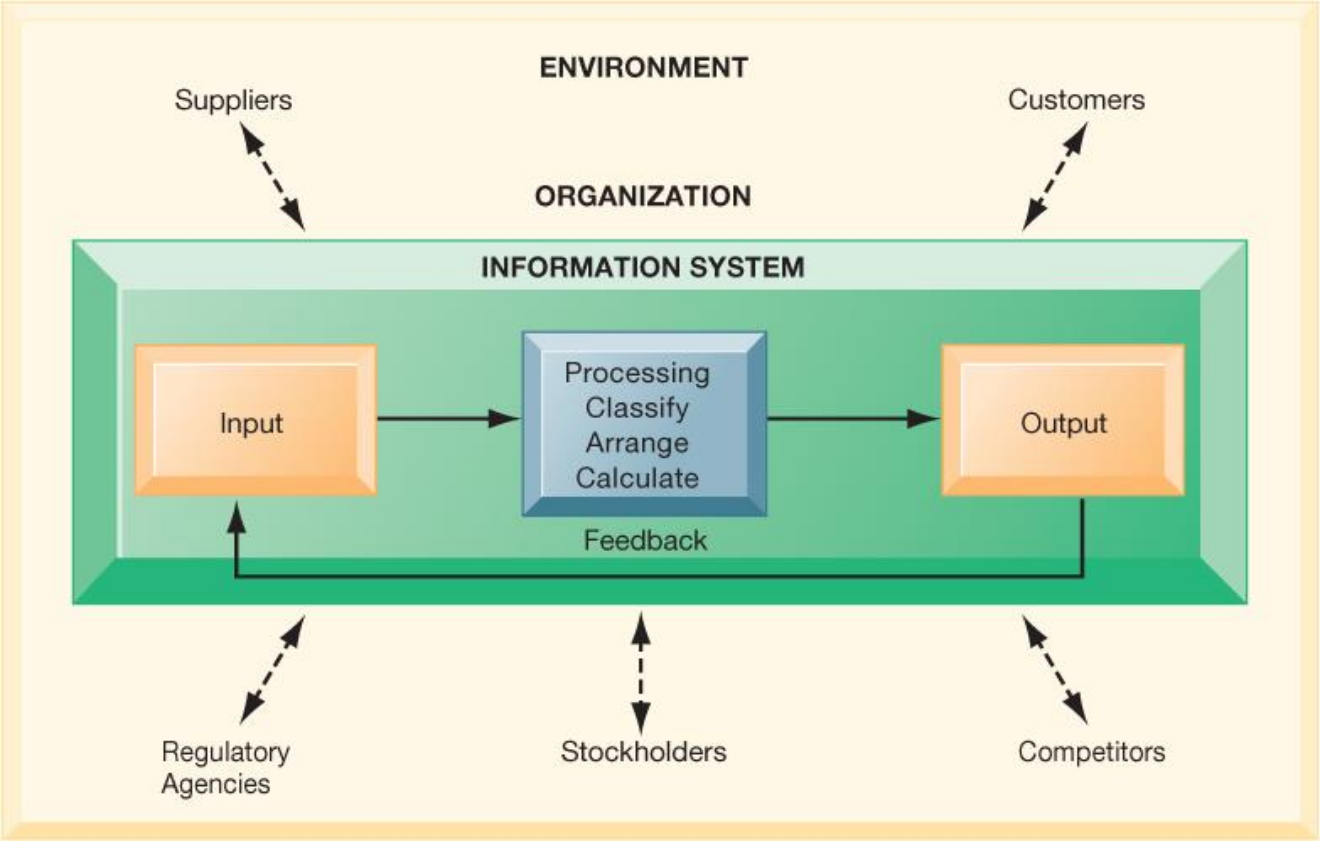
Data and Information

- **Data (or Raw Data)**
  - Streams of raw facts representing events occurring in organizations or the physical environment
- **Information**
  - Data shaped into meaningful form



Input, Processing, Output

- **Three activities of IS produce information organizations need**
  - Input, Processing, Output
  - Feedback: Output is returned to appropriate members of organization to help evaluate or correct input stage



Computers vs. Information System

- **Computer-based information systems use computer technology to transform data into information**
  - Computers and programs are the basic building blocks of information systems but don't create specific organizational information by themselves.
- **One must comprehend the problems they address, their design elements, and the organizational processes that lead to these solutions.**

Vs.

Image Source: Berkshire Hathaway, Eden Builders; Your Own Architect

Computer literacy vs. Information systems literacy

- **Computer/computer program vs. information system**
  - Computers and software are ***technical foundations and tools*** similar to the materials and tools used to build a house.

- **Computer literacy**
  - Primarily on knowledge of information technology
  - Limited to understanding how computer hardware and software work

- **Information systems literacy**
  - Comprehensive understanding of information systems
  - Includes a behavioral as well as a technical approach

Dimensions of Information Systems

1. Organizations
2. Management
3. Technology

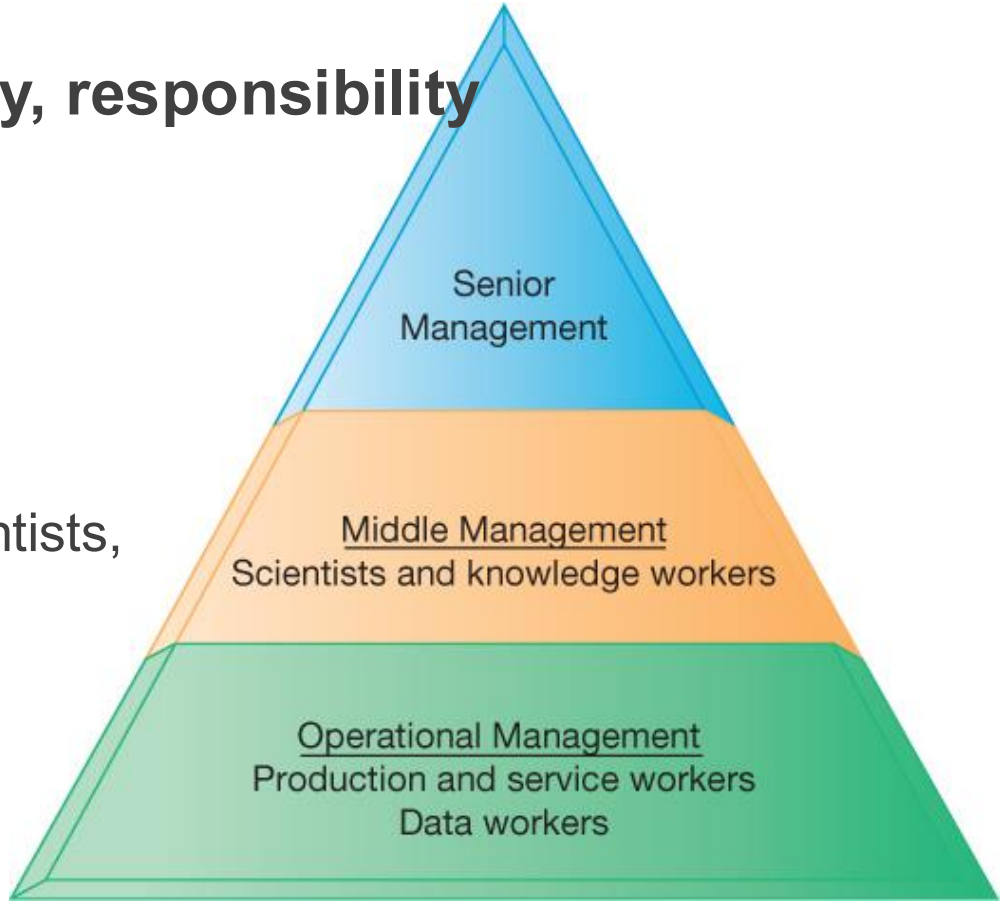
Organizations

1

- **Critical elements of an organization**
  - People, structure, business processes, politics, and culture.

- **[People] Hierarchy of authority, responsibility**
  - Senior management
  - Middle management
  - Operational management

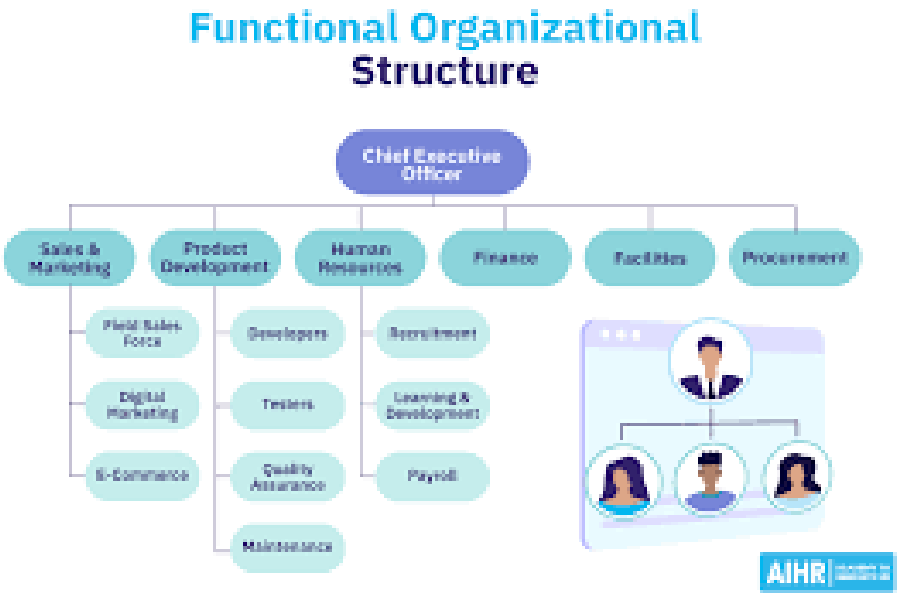
  - Knowledge workers (engineers, scientists, or architects)
  - Data workers (secretaries or clerks)
  - Production or service workers



Organizations

1

- **[Structure] Separation of business functions**
  - Sales and marketing
  - Human resources
  - Finance and accounting
  - Manufacturing and production



- **[Business processes]**
  - ***Formal rules*** that have been developed over a long time for accomplishing tasks
  - Some business processes are written down, but others are informal work practices
  - Information systems ***automate*** many business processes.

Image Source: AIHR

Organizations

1

- **Organizational [politics]**
  - Different levels and specialties in an org create different interests and points of view.
    - These views often ***conflict*** over how the company should be run and how resources and rewards should be distributed.
    - Conflict is the basis for organizational politics.
  - Information systems come out of this cauldron of differing perspectives, conflicts, compromises, and agreements that are a natural part of all organizations.
- **Unique business [culture]**
  - Fundamental set of assumptions, values, and ways of doing things, that has been accepted by most of its members.
  - Parts of an org's culture can always be found embedded in its information systems.
  - UPS's first priority is ***customer service***, which is an aspect of its organizational culture that can be found in the company's package tracking systems

Management

2

- **Management's job**
  - ***Making decisions***
  - Formulating ***strategy and action plans*** to solve organizational problems

- **In addition, managers must act *creatively***
  - Creation of new products and services
  - Occasionally re-creating the organization

- **Information technology can play a powerful role in helping managers design and deliver new products and services and redesigning their organizations.**

Technology

3

- **Computer hardware**
  - Physical equipment used for input, processing, and output activities in an IS
  - Computers of various sizes and shapes (including mobile handheld devices); various input, output, and storage devices; and telecommunications devices that link computers together
- **Computer software**
  - The detailed, preprogrammed instructions that control and coordinate the computer hardware components in an information system
- **Data management technology**
  - The software governing the organization of data on physical storage media
- **Networking and telecommunications technology**
  - Networks, the Internet, intranets and extranets, World Wide Web
- **IT infrastructure**
  - Platform that system is built on

A *Business Perspective* on Information Systems

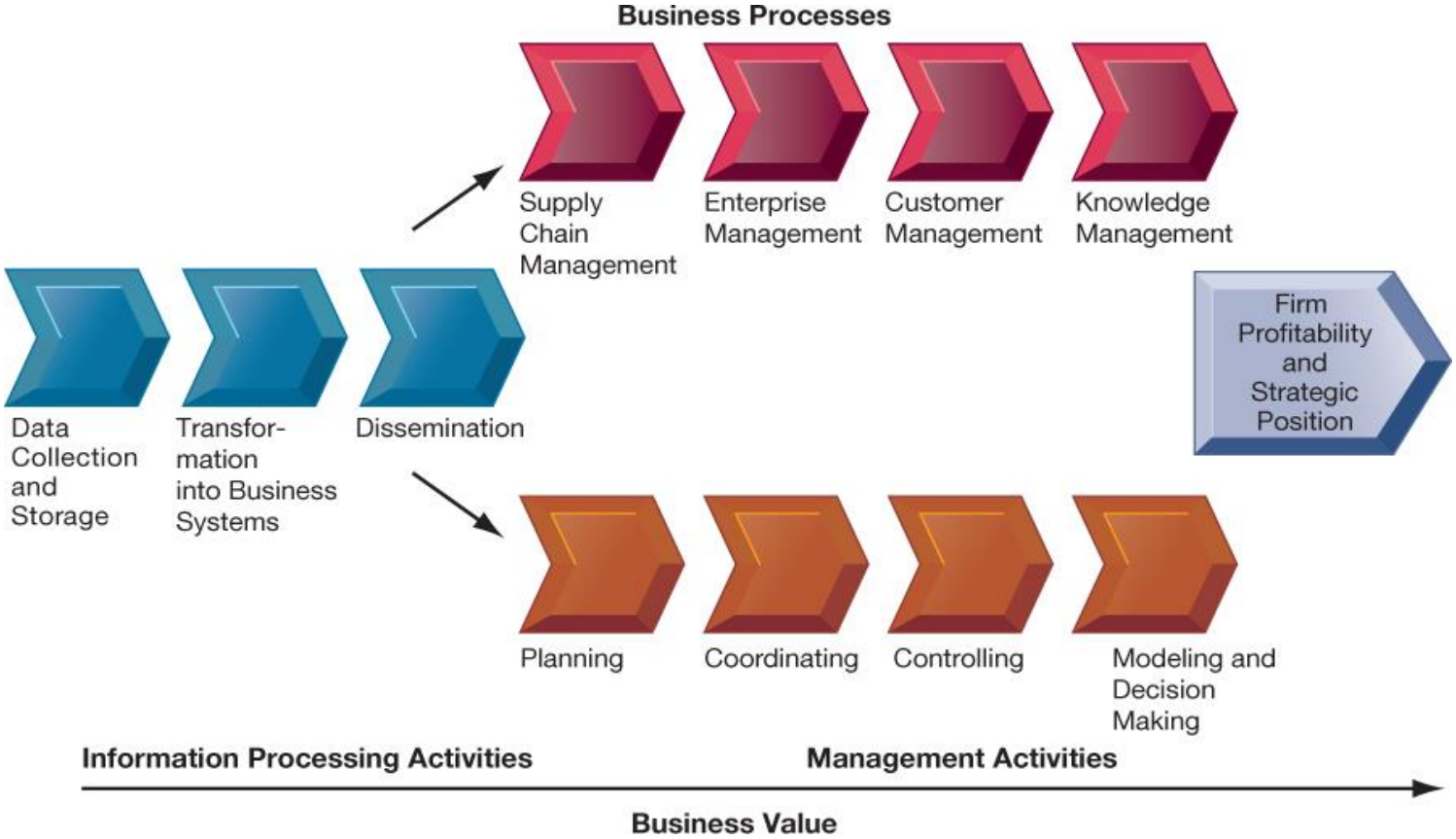
- **An information system is an instrument for decision-making**

- **Investments in information technology will result in superior returns**
  - Productivity and revenue increases
  - Superior long-term strategic positioning

- **Business perspective**
  - Calls attention to organizational and managerial nature of information systems

Business information value chain

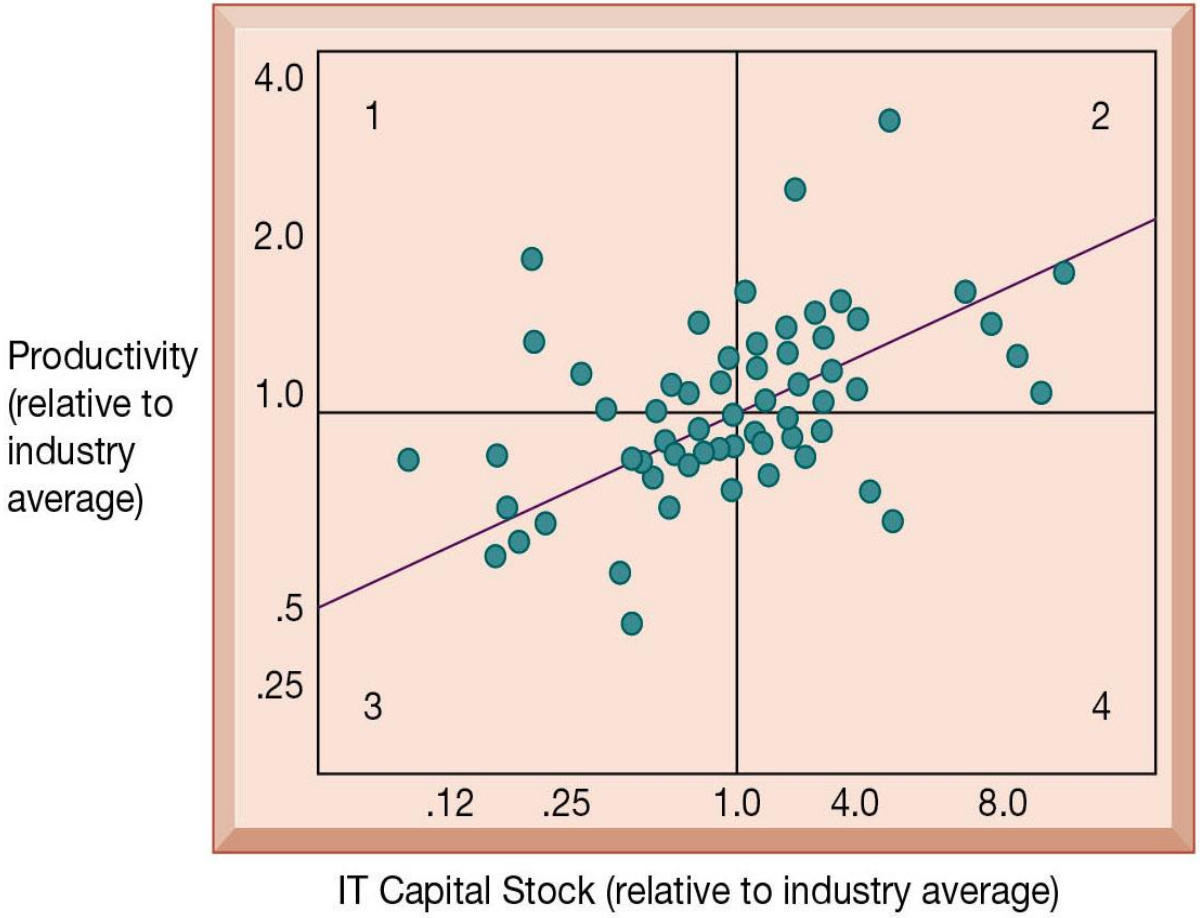
- **Raw data acquired and transformed through stages that add value to that information**
  - Value of information system determined in part by extent to which it leads to better decisions, greater efficiency, and higher profits



IT Investment ≠ Business Value

- **Investing in information technology does *not guarantee* good returns**
- **Considerable variation in the returns firms receive from systems investments**

What's the best case?



Complementary Assets

- ***Assets required to derive value*** **from a primary investment.**
  - e.g. Automobiles - highways, roads, gasoline stations, repair facilities, and a legal regulatory structure to set standards and control drivers

- **Research indicates that firms that support their technology investments with *investments in complementary assets* receive superior returns**
  - Example? ➔

- **These investments in organization and management are also known as *organizational and management capital*.**

Organizational, Managerial, and Social Assets

- **Complementary assets**
  - Examples of ***organizational*** assets
    - Appropriate business model
    - Efficient business processes
  - Examples of ***managerial*** assets
    - Incentives for management innovation
    - Teamwork and collaborative work environments
  - Examples of ***social*** assets
    - The Internet and telecommunications infrastructure
    - Technology standards

Organizational, Managerial, and Social Assets (Cont.)

Organizational assets	Supportive organizational culture that values efficiency and effectiveness Appropriate business model Efficient business processes Decentralized authority Distributed decision-making rights Strong IS development team
Managerial assets	Strong senior management support for technology investment and change Incentives for management innovation Teamwork and collaborative work environments Training programs to enhance management decision skills Management culture that values flexibility and knowledge-based decision making.
Social assets	The Internet and telecommunications infrastructure IT-enriched educational programs raising labor force computer literacy Standards (both government and private sector) Laws and regulations creating fair, stable market environments Technology and service firms in adjacent markets to assist implementation

1-3.

What academic disciplines are used to study information systems?

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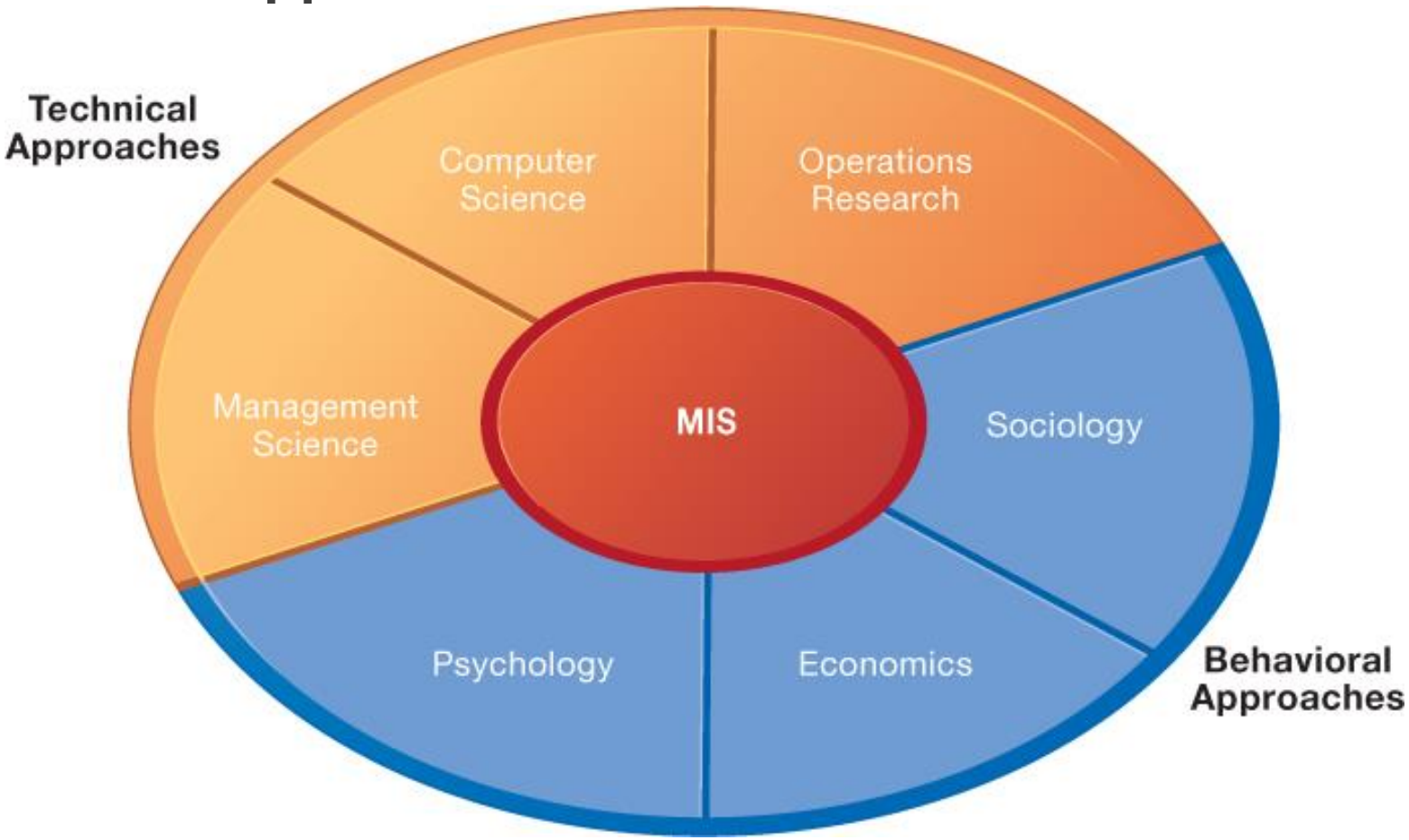
How does each contribute to an understanding of information systems?

Academic disciplines for MIS study

1. **Technical approaches**
2. **Behavioral approaches**



Contemporary Approaches to Information Systems

Technical Approach

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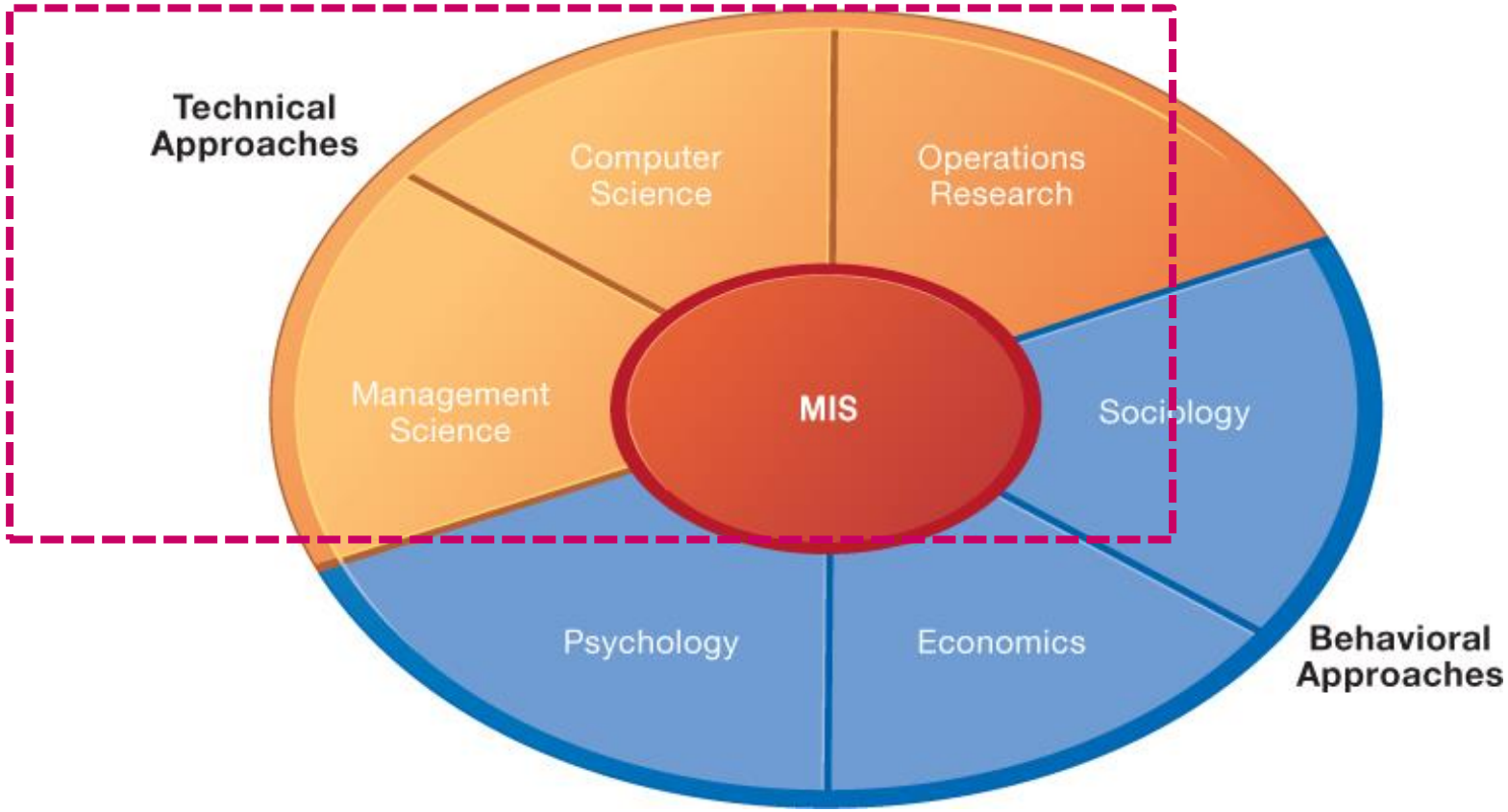
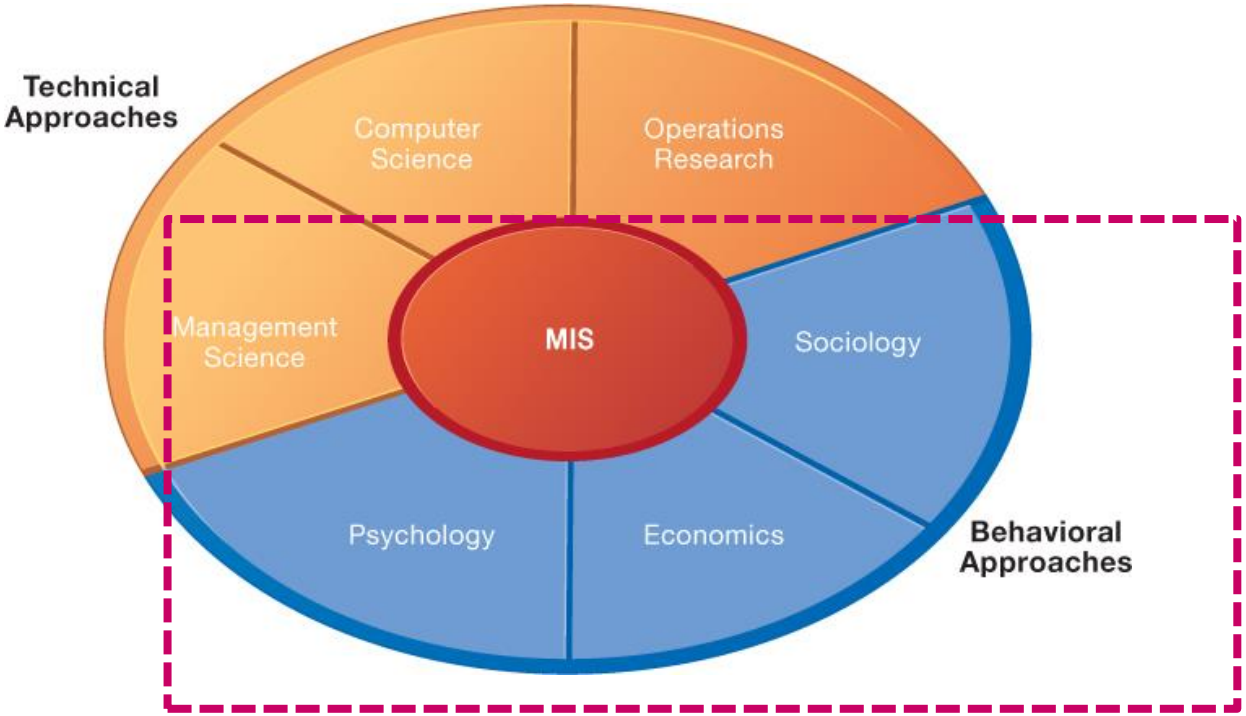
- **Emphasizes mathematically based models and the physical technology and formal capabilities of the systems.**
  - Computer science, management science, operations research



Behavioral Approach

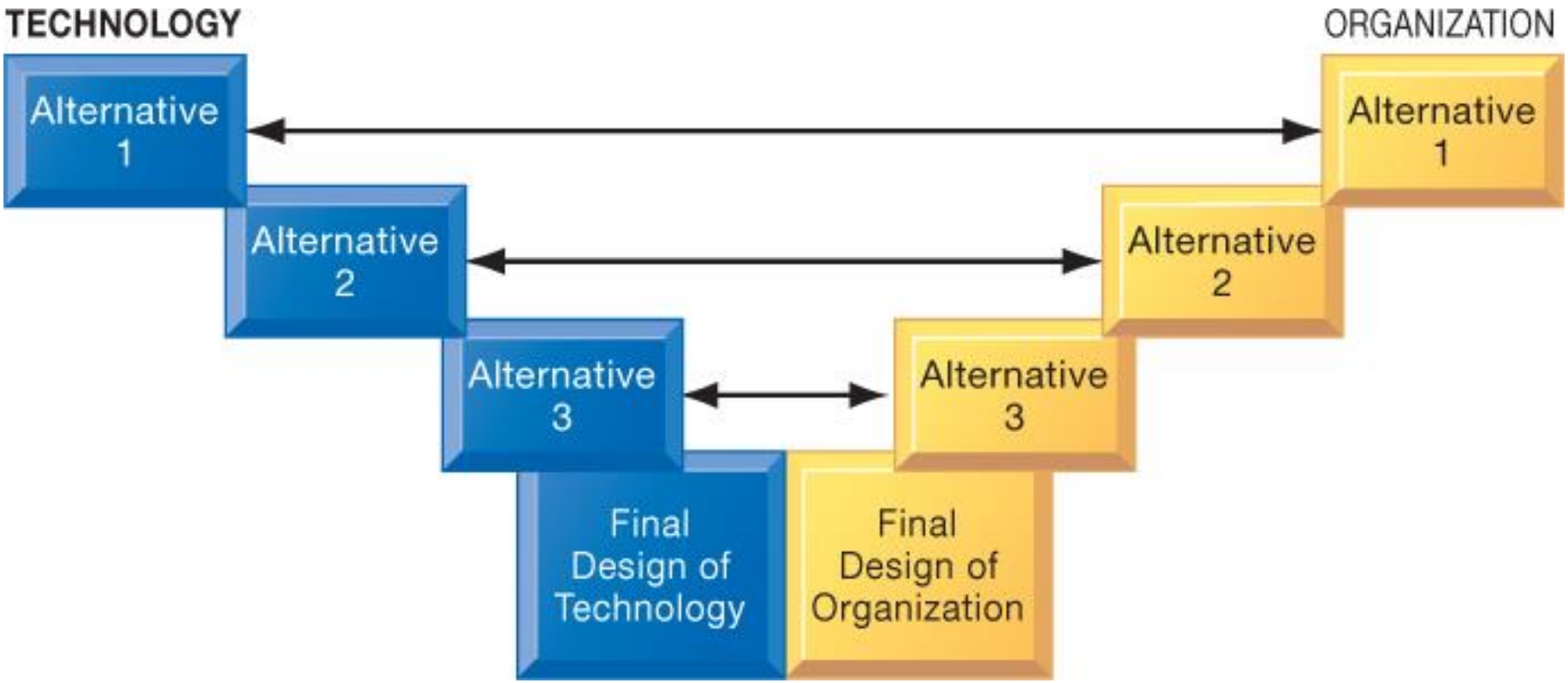
2

- **Concerned with behavioral issues that arise in the development and long-term maintenance of information systems (strategic business integration, design, implementation, utilization, etc.)**
  - Sociology, psychology, economics



Sociotechnical View

- Optimal organizational performance achieved ***by jointly optimizing*** both social & technical systems used in production
- Helps avoid purely technological approach



Key Takeaways

- Information systems are a core driver of modern business transformation, enabling efficiency, coordination, decision-making, and competitive advantage in today's digital economy.
- An information system is a socio-technical system, composed of management, organizational, and technological components that work together to support business processes and decisions.
- Technology alone does not create value; Complementary assets are essential for information systems to deliver real business benefits.
- Information systems are studied from multiple academic perspectives, each contributing unique insights into how systems are designed, used, and managed.

Next Week...

- **Next week, we will cover:**
  - [Chapter 2] Global E-business and Collaboration

Q & A